Zero-Shot Formation Classification in Amateur American Football

Lukas Brandner

08.10.26

SCIENCE
PASSION
TECHNOLOGY



Outline

1 Motivation

2 Background

3 Data & Methods

4 Results

5 Discussion & Conclusion

2026-10-07 Formation Classification

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Why automate formation tagging?

- American football is “chess on the field”: pre-snap formations carry tactical information coaches rely on
- At the amateur level (e.g. Austrian Football League), formation tagging on platforms like Hudl is done **entirely by hand**
- Open-weight Vision-Language Models (VLMs) can be run locally, could they help?

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Research Questions

RQ1 To what extent can user-grade open-source VLMs identify offensive formations from pre-snap images in a zero-shot setting?

RQ2 How do Qwen-VL, Gemma, and LLaMA compare in their ability to detect formations and player alignments?

RQ3 What are the main failure modes and limitations when applying VLMs to amateur-quality football broadcast footage?

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What a pre-snap formation shows

- Backfield occupied (RB/FB), or empty?
- How many wide receivers per side?
- Tight end/fullback on the line?
- Which side has the extra receiver (“strong side”)?

Why this is hard: precise counting and spatial reasoning

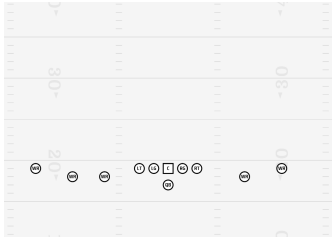
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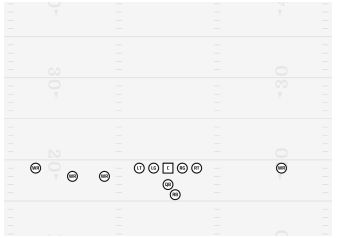
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- Four visual cues define a formation: backfield occupied or empty, WR count per side, TE/FB on the line or not, which side has the extra receiver ("strong side").
- Why it’s hard: needs precise counting + spatial reasoning — easy to miss even for a person on grainy broadcast footage.

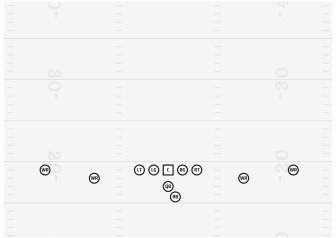
The Formation Classes (1/2)



EMPTY



TRIO

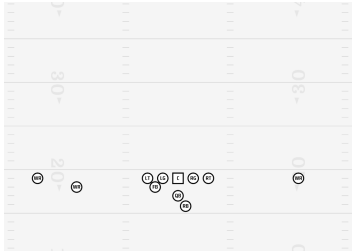


DOUBLES

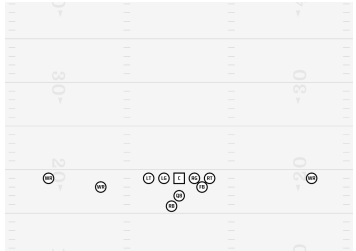
The Formation Classes (1/2)



The Formation Classes (2/2)



TREY



TANK

The Formation Classes (2/2)



Four Games, Four Different Cameras: Up Close



Game 1



Game 3

Four Games, Four Different Cameras: Up Close



Game 1 Game 3

Four Games, Four Different Cameras: From Further Away



Game 2



Game 4

Four Games, Four Different Cameras: From Further Away



Game 2 Game 4

Dataset

- 40 offensive plays, 4 AFL games
- One pre-snap frame per play
- Manual annotations: formation, play type, jersey colors, offense side
- Majority class: **TRIO** (35%)

Formation	Count	%
TRIO	14	35.0
DOUBLES	11	27.5
EMPTY	5	12.5
TREY	5	12.5
TANK	5	12.5
Total	40	100.0

- 40 offensive plays across 4 AFL games, one pre-snap frame per play.
- Manually annotated: formation, play type, jersey colors, offense side.
- Majority class TRIO at 35%

Models Under Test

Model	Params	Quantization	Disk size
Qwen3-VL:30b	30B	4-bit	19 GB
Gemma3:27b	27B	4-bit	17 GB
LLaMA3.2-Vision:11b	11B	4-bit	8 GB

- All served locally via **Ollama**, single 24 GB GPU
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Evaluation Protocol

Phase A Text-only: do models know the formation *names* at all, with no image?

Phase B Basic image understanding: jersey colors, offense side, player counts, QB alignment

Phase C Formation classification: the core task, in three prompting strategies

Majority-class baseline: always predict TRIO ⇒ **35.0% accuracy**

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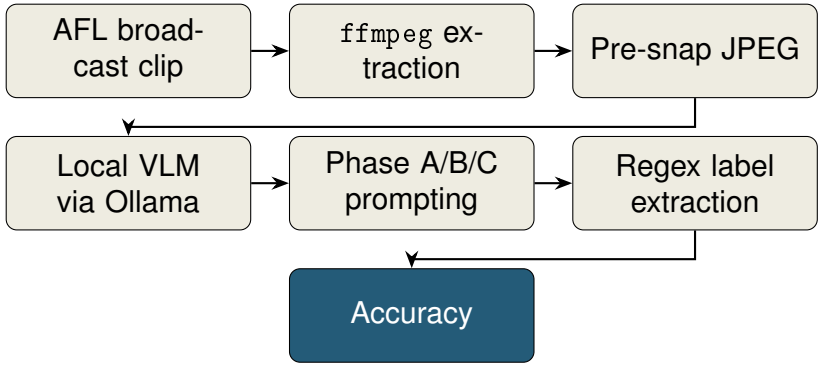
Phase C: Three Prompting Strategies

C1	C2	C3
Baseline: image + class names only	+ natural-language description per class	+ step-by-step decision tree

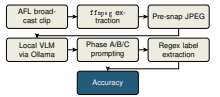
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Pipeline Overview



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Phase A: Inconsistencies, Before Any Image (1/2)

Q	Asked	Qwen3-VL	Gemma3	LLaMA3.2
A1	Describe each formation	Empty	Empty	Empty and Trey
A2	No RBs?	✓	✓	✓
A3	TEs or FBs?	×	✓	×
A4	3 true WRs?	×	×	×
A5	2 WRs each side?	✓	✓	×
A6	TRIO vs. TREY?	×	×	×
A7	DOUBLES vs. TANK?	×	×	×

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A5	2 WRs each side?	✓	✓	×
A6	TRIO vs. TREY?	×	×	×
A7	DOUBLES vs. TANK?	×	×	×

- Walk the table row by row:
- A2 (name EMPTY): all three correct.
- A3 (which formations use a TE/FB): only Gemma correct.
- A4 (name the 3-WR-one-side formation, i.e. TRIO): all three wrong. all had trey
- A5 (name DOUBLES): Qwen and Gemma correct, LLaMA wrong.
- A6 (TRIO vs. TREY, explain the difference): all three wrong.
- A7 (DOUBLES vs. TANK, explain the difference): all three wrong.

Phase A: Inconsistencies, Before Any Image (2/2)

Finding

EMPTY is standard football terminology: all three get it right. DOUBLES, TRIO, TREY, and TANK are not standardized: models struggle to consistently describe or distinguish them

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Phase B: Basic Image Understanding

- Sanity checks: jersey colors, offense side, player counts, QB positioning
- Are models even reliably reading the image (player counts, empty responses)?
- **Swap analysis**: of all offensive-jersey-color errors, how many are the model naming the *defense's* true color instead?
- Does giving the offense side explicitly fix this?

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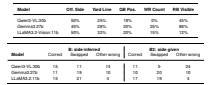
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Phase B: Results (1/2)

Model	Off. Side	Yard Line	QB Pos.	WR Count	RB Visible
Qwen3-VL:30b	50%	25%	18%	0%	45%
Gemma3:27b	45%	28%	20%	25%	88%
LLaMA3.2-Vision:11b	50%	32%	20%	15%	12%

	B: side inferred			B2: side given		
Model	Correct	Swapped	Other wrong	Correct	Swapped	Other wrong
Qwen3-VL:30b	15	11	14	11	5	24
Gemma3:27b	11	19	10	10	20	10
LLaMA3.2:11b	15	21	4	17	19	4

Phase B: Results (1/2)



Phase B: Results (2/2)

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Basic image reading is inconsistent, and often not really “reading” at all.

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Code Walkthrough

- Live in `evaluation.ipynb`: Ollama API call and prompts C1 $\rightarrow$ C3

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Phase C: Formation Classification Accuracy

Model	C1	C2	C3
Qwen3-VL:30b	10.0%±9	30.0%±14	27.5%±14
Gemma3:27b	15.0%±11	**37.5%±15**	12.5%±10
LLaMA3.2-Vision:11b	12.5%±10	12.5%±10	20.0%±12
Majority baseline		**35.0%**	

± = 95% CI (binomial approx., *n*=40 per cell) – most pairwise differences between phases and models overlap within noise.

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Phase C: Models Default, They Don’t Classify

Formation	Ground truth	Predicted	Ratio
TRIO	35.0%	15.8%	0.5×
DOUBLES	27.5%	10.0%	0.4×
EMPTY	12.5%	1.1%	0.1×
TREY	12.5%	61.9%	**5.0**×
TANK	12.5%	11.1%	0.9×

Across all 360 Phase C predictions (3 models × 3 phases × 40 images): TREY is guessed 5× more often than it’s actually correct, while EMPTY is almost never guessed at all - a favourite default, not a per-image signal.

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Why It’s Not Working (2/2): Perception

- Even the basic “read the image” step mostly fails, before formation logic can even start
- Identifying which side is even the offense: 45-50%
- Wide-receiver count: 0% (Qwen), 15% (LLaMA), 25% (Gemma: comes from reciting “3” on every single image, not counting)
- Running back in the backfield: LLaMA answers “No”, Gemma always “Yes”

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Limitations

- Small test set: 40 images, minority classes have only 5 samples each
- Single competition (AFL): may not generalize to other leagues/cameras
- All models run at 4-bit quantization, constrained by 24 GB VRAM
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RQ2 No model shows real multi-class discrimination. Peak accuracy doesn't track parameter count (30B and 27B tie at 37.5%, 11B close behind).

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